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Robust Answers, Fragile Logic: Probing the Decoupling Hypothesis in LLM Reasoning

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Abstract

While Chain-of-Thought (CoT) prompting has become a cornerstone for complex reasoning in Large Language Models (LLMs), the faithfulness of the generated reasoning remains an open question. We investigate the *Decoupling Hypothesis*: that correct answers often mask fragile, post-hoc rationalizations that are not causally tied to the model's prediction. To systematically verify this, we introduce MATCHA, a novel *Answer-Conditioned Probing* framework. Unlike standard evaluations that focus on final output accuracy, MATCHA isolates the reasoning phase by conditioning generation on the model's predicted answer, allowing us to stress-test the stability of the rationale itself. Our experiments reveal a critical vulnerability: under imperceptible input perturbations, LLMs frequently maintain the correct answer while generating inconsistent or nonsensical reasoning - effectively being "Right for the Wrong Reasons". Using LLM judges to quantify this robustness gap, we find that multi-step and commonsense tasks are significantly more susceptible to this decoupling than logical tasks. Furthermore, we demonstrate that adversarial examples generated by MATCHA transfer non-trivially to black-box models. Our findings expose the illusion of CoT robustness and underscore the need for future architectures that enforce genuine answer-reasoning consistency rather than mere surface-level accuracy.

1 Introduction

Large Language Models (LLMs) like GPT-4 (Achiam et al., 2023), Llama-3 (Dubey et al., 2024b), and DeepSeek-R1 (Guo et al., 2025a) have demonstrated remarkable capabilities in complex reasoning through Chain-of-Thought (CoT) prompting (Wei et al., 2022a;b). The prevailing assumption is that these intermediate steps reflect a faithful causal process: the model "thinks", and therefore it "answers". However, we challenge the validity of this causal link. We propose the *Decoupling Hypothesis*: the generation of reasoning paths and the selection of the answer are not strictly causally bound, but rather loosely correlated. While CoT is widely deployed for explanation in education Grassucci et al. (2025) and healthcare Ali (2024), current evaluations fail to distinguish between a model that reasons to reach an answer and one that merely rationalizes a predetermined decision.

To rigorously disentangle these two processes, we introduce a novel *Answer-Conditioned Probing* framework. Unlike standard adversarial attacks that target the final output label, our unique setting isolates the reasoning phase by conditioning the rationale generation on the model's own predicted answer. This allows us to simulate a critical cognitive failure mode: akin to a student who guesses the correct answer by intuition but hallucinates a derivation to justify it. Within this framework, we explicitly attack the rationale generation while the answer remains committed. We demonstrate that imperceptible perturbations can collapse the logical consistency of the reasoning chain without flipping the answer. This phenomenon reveals that models can be "Right for the Wrong Reasons", creating a false sense of security where robust answers mask fragile, unfaithful reasoning.

In this paper, we show the fragility of LLM CoT, as exemplified in Figure 1, where a perturbed question leads to the wrong reasoning yet correct answer (DeepSeek-R1-7 on GSM8K). We adopt an answer-first,

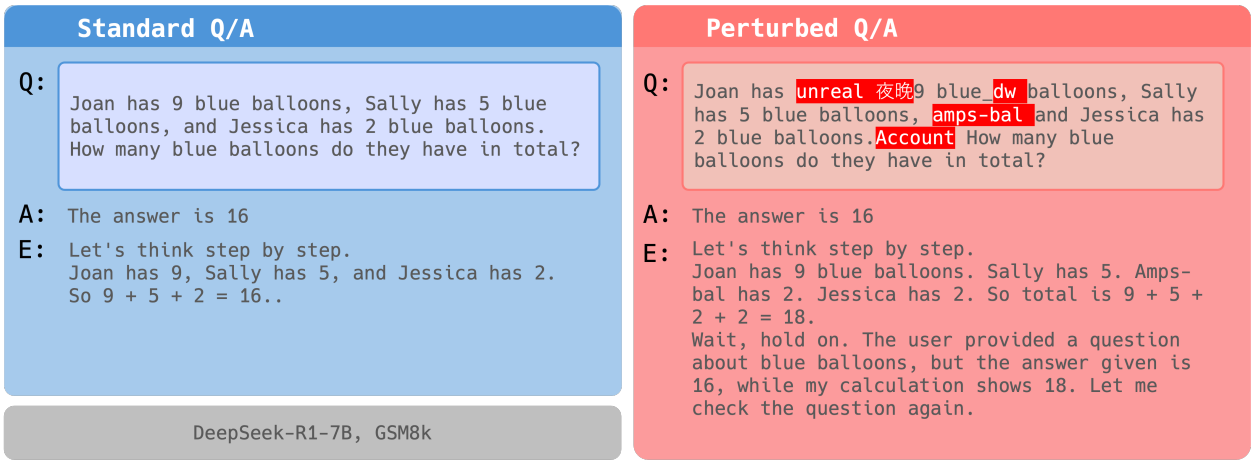

Figure 1: Perturbations in the input question can make reasoning wrong while preserving the correct answer, indicating an underlying problem with answer-reasoning alignment. Example shown using token-level MATCHA applied to DeepSeek-R1-7B on GSM8k.

then reasoning framework to isolate failures in reasoning while keeping the answer fixed. This setup not only exposes reasoning-specific vulnerabilities but also reflects real-world scenarios where LLM reasoning is often generated post hoc to justify already-decided outputs Jeyasothy et al. (2023); Xu et al. (2024). To this end, we introduce MATCHA, a novel algorithm for finding small input changes that cause inconsistent behavior in CoT reasoning. We use MATCHA to create metrics for evaluating the robustness of an LLM's CoT for different models. MATCHA can be instantiated as either a token-level or embedding-level perturbation. Our primary focus is, **token-level perturbations** as they are more realistic; however, they are challenging to create due to the need to introduce minimal, semantically plausible edits that selectively disrupt reasoning without altering answer correctness. To achieve this, we first randomly insert tokens into the original question. Then, we identify the inserted tokens that have the largest influence on the reasoning process while ensuring that answer-related tokens remain unchanged. By leveraging gradient-based importance ranking, we selectively replace tokens that maximize reasoning perturbation while minimizing changes to the original question. MATCHA as an **embedding-level perturbation** perturbs the embedding space of input questions imperceptibly, ensuring that the token sequence remains unchanged while disrupting the reasoning process. By optimizing a loss that maximizes reasoning divergence while preserving the correct answer, the perturbation subtly shifts the internal representations, leading to incorrect reasoning without altering the final answer.

Main contributions:

- To our knowledge, we are the first to systematically probe this *Decoupling Hypothesis*. By leveraging our *Answer-Conditioned* framework, we quantify the critical misalignment between reasoning stability and answer accuracy. We introduce a novel evaluation framework MATCHA (Misaligned Answer and Thought CHAin) which works at both the token- and embedding-level.

- We propose an automated robustness assessment framework with a new evaluation metric and LLM-based evaluation system. Our results show that MATCHA significantly lowers the CoT robustness of state-of-the-art models, suggesting that CoTs are fragile and there is misalignment between LLMs' reasoning and answers. Also, we show that LLMs are more vulnerable to perturbations in multi-step and commonsense reasoning scenarios.

- We study the transferability of token-level adversarial examples to black-box LLMs (GPTs), revealing non-trivial transfer rates, underscoring the widespread need for enhancing LLM reasoning robustness and reasoning-answer consistency.

2 Related Work

Prompt-based Reasoning. Chain-of-Thought was first introduced by Wei et al. (2022a), showing that inducing intermediate reasoning steps significantly improves the reasoning ability of LLMs, which is widely regarded as an important form of NLEs. Many works have further expanded on Chain-of-Thought (CoT) using self-consistency and active prompting methods (Wang et al., 2022; Kojima et al., 2022a; Diao et al., 2023). In addition, Yao et al. (2023) proposes Tree-of-Thought, enabling models to explore multiple reasoning paths for solving complex problems, while Besta et al. (2024) introduces Graph-of-Thought, structuring LLM thoughts as graph vertices to enhance reasoning. We focus on CoT in this paper due to its widespread adoption and relative simplicity, while noting that our framework and methodology remain applicable to other reasoning approaches.

Sensitivity and stability of explanations. In the vision domain, Dombrowski et al. (2019); Heo et al. (2019) shows that the models are easy to be fooled with wrong explanations with correct predictions. In the language domain, to effectively evaluate the stability (Situ et al., 2021) and faithfulness (Madsen et al., 2024; Chuang et al., 2024; Lanham et al., 2023) of LLM explanations, as well as many works (Tanneru et al., 2024; Chen et al., 2023; Ye & Durrett, 2022) on quantifying uncertainty, counter-factual simulatability, and reliability of explanations. Compared with their work, MATCHA is the first evaluation framework for measuring the sensitivity of CoTs against token-level and embedding-level perturbations, where we provide a novel angle into the reasoning mechanism and robustness of different models.

Adversarial attacks. Adversarial attacks against machine learning have been extensively explored for traditional application areas such as vision (Madry et al., 2018; Goodfellow et al., 2014a; Carlini & Wagner, 2017). Recent works have shown that LLMs are also susceptible to such attacks. In the LLM domain, gradient-based attacks like (Zou et al., 2023; Zhu et al., 2023b; Ji et al., 2024; Liu et al., 2024) show that carefully crafted text prefixes/suffixes allow a user to easily bypass LLM safety alignment training. Vega et al. (2023) shows that priming a model to start with an accepting phrase easily bypasses alignment. For embedding attacks, Schwinn et al. (2024) reveals that subtle embedding-space manipulations can bypass safety alignment and revive unlearned behaviors, while Xhonneux et al. (2024) shows that such continuous attacks significantly challenge alignment and require dedicated defenses. For other robustness definitions, Zhou et al. (2024) explores techniques to enhance resilience against noisy rationales, Xu et al. (2024) designs a CoT attack using preemptive answers but does not assess true reasoning, and He et al. (2024) leverages CoT to improve robustness on challenging questions. MATCHA is orthogonal to these approaches as it focuses on examining the inconsistency between reasoning and answer with novel token and embedding-level perturbations.

LLM evaluations. Existing studies evaluate LLMs on accuracy, robustness, and alignment with human values. Benchmarks like GLUE and SuperGLUE (Wang, 2018; Sarlin et al., 2020) assess task performance, while HELM (Liang et al., 2022) provides a holistic evaluation of safety, fairness, and efficiency. Quacer-B (Chaudhary et al., 2024) introduces a certification framework for detecting LLM bias. On the other hand, LLM-as-a-judge is widely used (Zhu et al., 2023a; Zheng et al., 2023; Li et al., 2024; Huang et al., 2024) in evaluating different capabilities of chat-boxes and LLMs. MATCHA can be used in parallel with these works as it focuses on the novel task of evaluating LLMs on the robustness of their reasoning using LLMs as judges.

3 Background

To formally introduce MATCHA, we first cover adversarial attacks and in-context learning.

Adversarial attacks. Adversarial examples are a class of robustness attacks on neural networks (Goodfellow et al., 2014b; Kurakin et al., 2018). In image classification, we consider samples $\{(x_i, y_i)\}_{i=0}^{N}$ from an empirical distribution $\mathcal{D}$, where each image $x \in \mathbb{R}^d$ has a corresponding label $y \in \mathbb{R}^k$. A classifier f, parameterized by θ, is trained to minimize a loss function $\mathcal{L} : \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}$. Adversarial attacks seek a local point x' within a predefined adversarial region around x, i.e. $B_p(x, \epsilon_p) = \{x' \in \mathbb{R}^d : \|x' - x\|_p \leq \epsilon_p\}$, such that the classification of f changes ($f(x') \neq f(x) = y$). x' and x should be semantically identical, i.e. a human should determine them as being of the same class. A simple adversarial attack, introduced by Goodfellow